Module 5: Feature Selection & Skewness Report

1. Feature Removal Justification

To optimize machine learning performance and prevent distance calculation distortions in K-Means clustering, low-information and redundant columns were systematically pruned:

Removed Feature	Pruning Reason / Category	Impact on ML Pipeline
ID	Unique Key / Arbitrary Identifier	Prevents overfitting and random cluster partitioning.
Z_CostContact & Z_Revenue	Constant Values (Zero Variance)	Eliminates dead attributes that add zero variance information.
Year_Birth & Dt_Customer	Redundant Temporal Data	Replaced by standardized *Customer_Age* and *Customer_Tenure*.
Marital_Status & Education	Raw Categorical Strings	Replaced by One-Hot Encoded sparse binary matrices.

2. Skewness Assessment & Log Transformations

K-Means clustering relies on Euclidean distance calculations. Skewed financial variables with extreme long-tail distributions distort distance matrices, leading to imbalanced cluster allocations.

Applied Transformation Formula:

$$x_{transformed} = \ln(1 + \max(0, x))$$

The log1p transformation was selected to smoothly compress right-skewed revenue variables while safely handling zero purchase values.

Variable Name	Pre-Transform Skew	Transformation Applied	Post-Transform Status
Total_Spending	High Right Skew (> 1.5)	Log Transformation (log1p)	Normalized / Gaussian-like
Income	Severe Skew & Outliers	Median Imputation + log1p	Stabilized variance
Avg_Spending_Per_Purchase	Moderate Right Skew	Log Transformation (log1p)	Balanced Distribution